



SHR4_SW D6_Team 1



Full-Cycle Software Testing Project

Team Members:

- 1- Faris Nabil
 - 2- Ammar Mohammed
 - 3- Gody Sherif
 - 4- Alaa Emad
 - 5- Mariam Tarek
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Project Idea Summary:

This project aims to apply the complete Software Testing Life Cycle (STLC) on a real-world application to ensure software quality, reliability, and performance. The project covers all testing phases, starting from requirements analysis and test planning, followed by designing user stories, creating detailed test cases, executing manual testing, and reporting defects.

The project also includes API testing using Postman, mobile application testing, and automation testing using Selenium WebDriver with Java, TestNG, and Maven. End-to-End testing scenarios were implemented to validate the application's critical business workflows. In addition, project management artifacts such as the Gantt Chart and Network Diagram were prepared to organize and monitor project activities.

The main objective of the project is to verify that the system meets user requirements, identify defects, improve software quality, and demonstrate practical skills in manual, automated, mobile, and API testing within a complete quality assurance process.

Project Description

E-Commerce Systems

- This project focuses on performing a complete Software Testing Life Cycle (STLC) for a real-world E-Commerce application. The objective was to ensure the quality, reliability, security, and usability of the system through comprehensive testing activities covering web, mobile, API, and automated testing.**
- The application under test serves multiple user roles, including Customers, Administrators, and**

Brands/Sellers. The system provides various e-commerce functionalities such as user registration and authentication, multi-factor authentication (MFA), product browsing, category filtering, shopping cart management, checkout and payment processing, customer profile management, invoices, favorites, messaging, and discount management.

- **The project began with requirement analysis and test planning, followed by the creation of 19 User Stories representing the major business functionalities of the system. Detailed test scenarios and test cases were designed for each user story, covering positive, negative, boundary, integration, and user interface testing techniques.**
- **Manual testing activities included component testing, integration testing, UI testing, functional testing, and end-to-end testing. Hundreds of test cases were executed across different modules, resulting in the identification and documentation of multiple software defects through structured bug reports based on severity and priority levels.**
- **API Testing was conducted using Postman to validate REST API endpoints, ensuring correct status codes, response structures, authorization mechanisms, input validation, and overall API reliability. CRUD operations (GET, POST, PUT, and DELETE) were tested to verify proper backend functionality.**

- **Automation Testing was implemented using Selenium WebDriver with Java, TestNG, and Maven. The automation framework followed industry-standard practices and included automated end-to-end scenarios covering critical business workflows such as registration, login, shopping cart operations, and checkout processes. Automated test execution and reporting improved testing efficiency and repeatability.**
- **Mobile Application Testing was performed on both Android and iOS platforms using the provided APK application. Functional testing, usability testing, and defect reporting activities were conducted to ensure a consistent user experience across mobile devices.**
- **Project planning and management activities were supported using a Gantt Chart covering an eight-week schedule and a Network Diagram illustrating project dependencies and workflow sequencing.**
- **The project successfully demonstrated the practical application of Quality Assurance methodologies, Manual Testing, API Testing, Mobile Testing, Automation Testing, Defect Management, and Test Documentation processes within a complete software testing environment.**

Problem Statement:

- **This project aims to ensure the quality, reliability, and security of an E-Commerce application by identifying defects before they impact end users. Through manual, API, mobile, and automation testing, the project helps deliver a smooth and seamless user experience. Security testing techniques, including validation against vulnerabilities such as SQL Injection, were performed to reduce security risks and protect user data.**
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Testing Environment

1- Devices Tested:

- **HP ZBook 15 G5**
- **ASUS TUF Gaming A15**
- **HP EliteBook**
- **Lenovo ThinkPad T470s**
- **Tecno Camon 20**
- **Redmi 9**
- **Redmi Note 14**
- **iPhone 13**

2- Operating Systems:

- **Windows 11 Pro**
- **Windows 10**
- **Android, iOS**